

# **AI-POWERED BUSINESS REQUEST ROUTING AUTOMATION**

A demonstration project for Borealis Consumer Electronics showcasing intelligent workflow automation using Google Forms, n8n, and OpenAI for automated internal request classification and routing



# THE BUSINESS CHALLENGE

## CURRENT PAIN POINTS

- Manual request routing creates bottlenecks
- Routing errors delay response times
- No consistent classification standards
- High-volume periods overwhelm teams
- Difficult to track request patterns

## IMPACT ON OPERATIONS

Without automated classification, requests require manual review before routing, creating delays and inconsistent handling. Teams spend valuable time on triage rather than resolution.

# INTELLIGENT SOLUTION OVERVIEW

An automated workflow combining Google Forms, n8n, and OpenAI to classify and route internal business requests with confidence-based validation

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## REQUEST INTAKE

Google Forms captures structured employee requests

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## DATA CAPTURE

Google Sheets stores all request data securely

03

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## AI CLASSIFICATION

OpenAI model analyzes and categorizes requests

04

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## VALIDATION

Confidence threshold ensures reliable routing

05

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## SMART ROUTING

High-confidence requests auto-route instantly

06

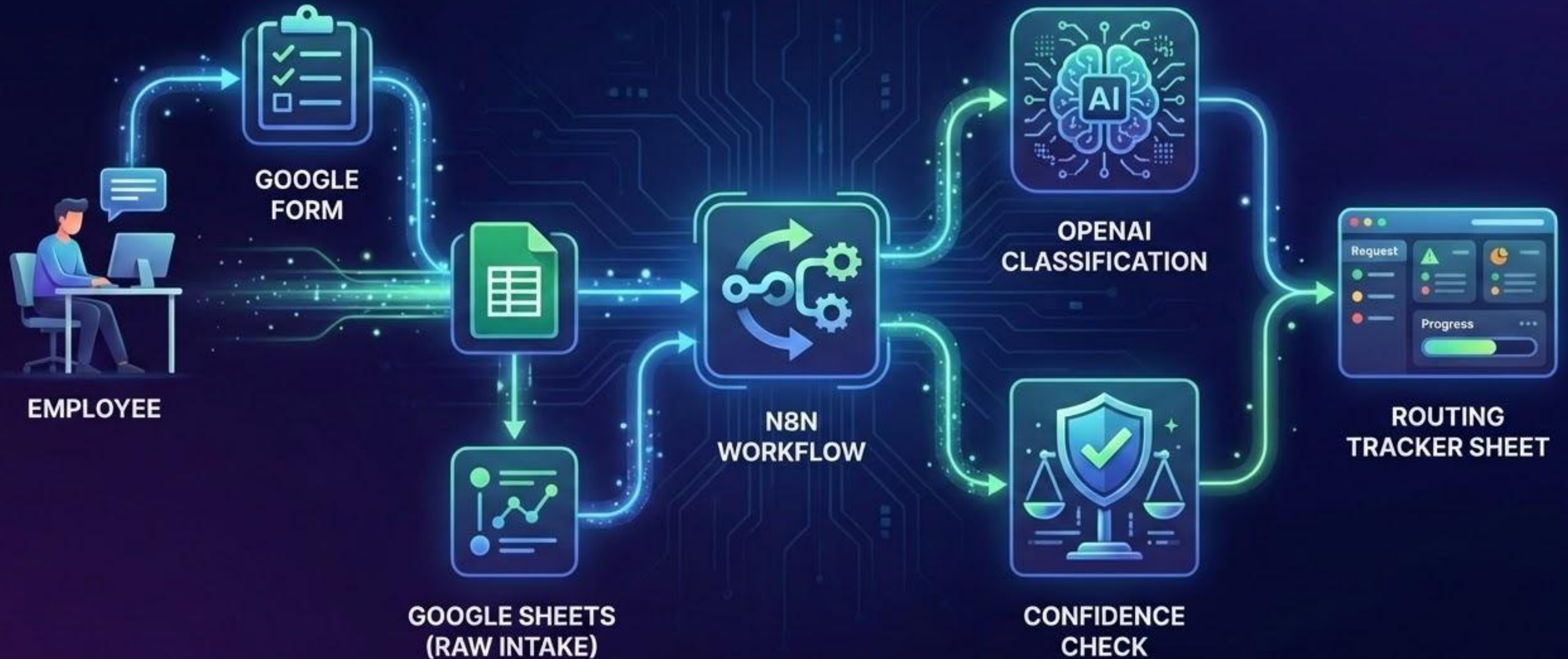
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## TRACKING

Complete audit trail with decision logging



# SYSTEM ARCHITECTURE



# AI CLASSIFICATION ENGINE

## INTELLIGENT REQUEST ANALYSIS

The OpenAI model processes request text using prompt engineering to extract three key attributes: request type (technical, HR, finance, facilities, or other), priority level (low, medium, or high), and target department.

The model returns structured JSON output that the workflow can parse and validate before routing decisions.

### Example LLM Classification Prompt:

*You are a business request classification assistant.*

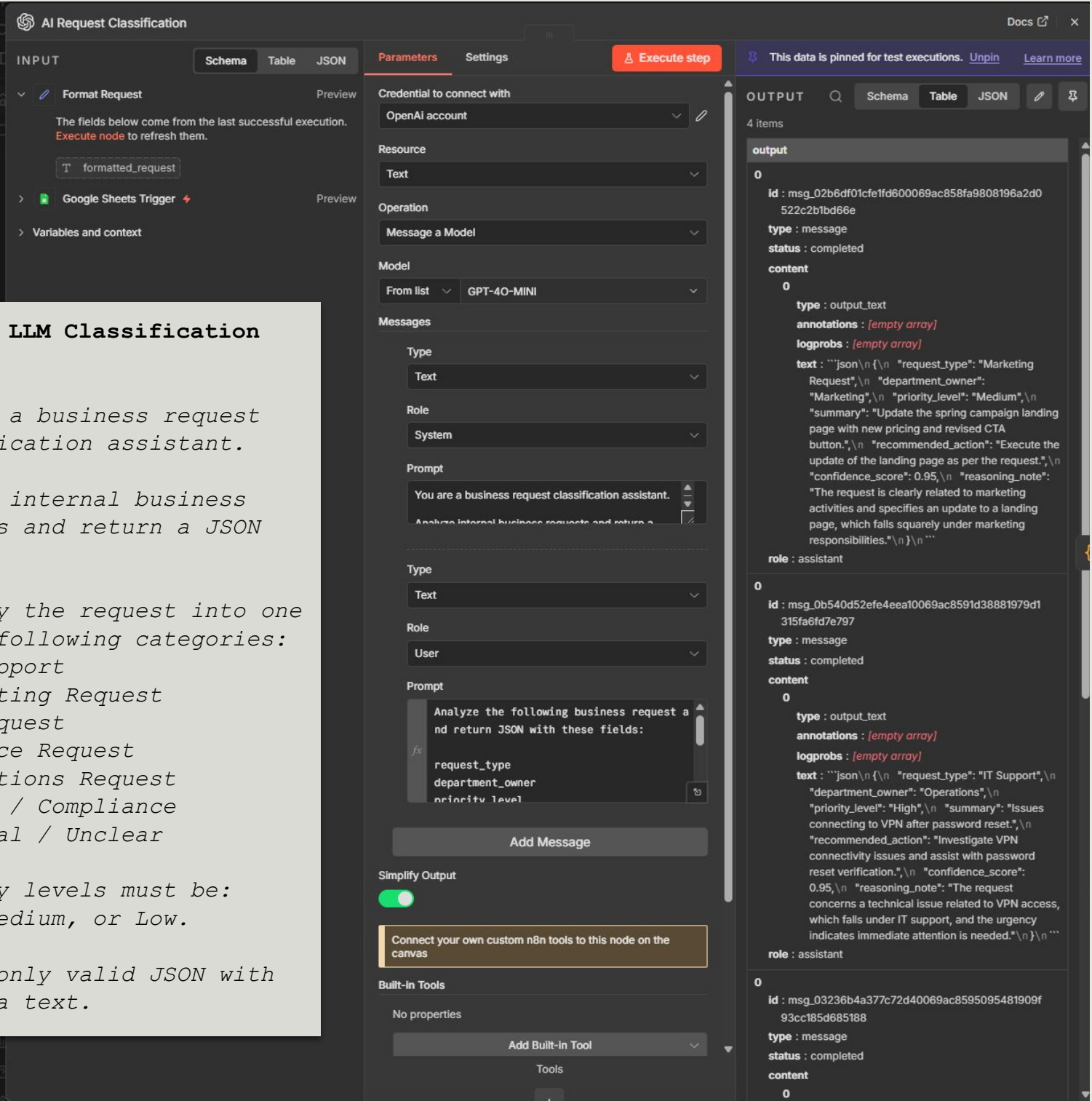
*Analyze internal business requests and return a JSON object.*

*Classify the request into one of the following categories:*

- IT Support
- Marketing Request
- HR Request
- Finance Request
- Operations Request
- Legal / Compliance
- General / Unclear

*Priority levels must be: High, Medium, or Low.*

*Return only valid JSON with no extra text.*





# VALIDATION LOGIC

Confidence-based threshold ensures routing reliability  
while maintaining human oversight for uncertain  
classifications

ParametersSettingsExecute step

Conditions

fx

{{\$.json.confidence\_score}}

# is greater than or equal to

0.75

+ Add condition

Convert types where required

Options

No properties

Add option

OUTPUTSchemaTableJSON

Borealis Operations Request System

File Edit View Insert Format Data Tools Extensions Help

Q Menus100%\$%123Default...10BIZAA

G18fx

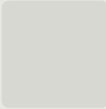
|    | A                             | B                  | C          | D        | E          | F                                                                                                                                                        |
|----|-------------------------------|--------------------|------------|----------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Table2                        |                    |            |          |            |                                                                                                                                                          |
| 1  | Timestamp                     | Request Type       | Owner      | Priority | Confidence | Reasoning Note                                                                                                                                           |
| 2  | 2026-03-07T15:43:29.529-05:00 | Marketing Request  | Marketing  | Medium   | 0.95       | The request is clearly related to marketing activities and specifies an update to a landing page, which falls squarely under marketing responsibilities. |
| 3  | 2026-03-07T15:43:29.530-05:00 | IT Support         | Operations | High     | 0.95       | The request concerns a technical issue related to VPN access, which falls under IT support, and the urgency indicates immediate attention is needed.     |
| 4  | 2026-03-07T15:43:29.531-05:00 | HR Request         | HR         | High     | 0.95       | The request is clear and urgent as it pertains to enabling a new employee's access prior to their start date.                                            |
| 5  | 2026-03-07T15:43:29.532-05:00 | Finance Request    | Finance    | Low      | 0.9        | The request is specifically related to finance operations, focusing on invoice processing and approval status.                                           |
| 6  | 2026-03-07T15:49:05.879-05:00 | Operations Request | Operations | Medium   | 0.85       | The request involves data accuracy within the dashboard, directly related to operations.                                                                 |
| 7  | 2026-03-07T15:50:08.733-05:00 | General / Unclear  | General    | Low      | 0.7        | The request is not specific to any department and lacks immediate urgency, making it a general inquiry about improving processes.                        |
| 8  | 2026-03-07T15:50:08.735-05:00 | IT Support         | HR         | Medium   | 0.85       | The request relates to technical access problems, which typically fall under IT support, even though initiated by HR.                                    |
| 9  | 2026-03-07T15:51:04.716-05:00 | Operations Request | Operations | Low      | 0.85       | The request involves a reporting issue within the operations department, indicating a need for operational review.                                       |
| 10 | 2026-03-07T15:54:05.630-05:00 | IT Support         | Finance    | Medium   | 0.85       | The request involves access issues related to IT systems and the potential inaccuracy of data, which falls under IT Support.                             |
| 11 | 2026-03-08T16:22:33.855-04:00 | Marketing Request  | Marketing  | High     | 0.95       | The request pertains specifically to marketing activities and is time-sensitive due to an upcoming campaign launch.                                      |
| 12 |                               |                    |            |          |            |                                                                                                                                                          |
| 13 |                               |                    |            |          |            |                                                                                                                                                          |
| 14 |                               |                    |            |          |            |                                                                                                                                                          |
| 15 |                               |                    |            |          |            |                                                                                                                                                          |
| 16 |                               |                    |            |          |            |                                                                                                                                                          |
| 17 |                               |                    |            |          |            |                                                                                                                                                          |
| 18 |                               |                    |            |          |            |                                                                                                                                                          |
| 19 |                               |                    |            |          |            |                                                                                                                                                          |
| 20 |                               |                    |            |          |            |                                                                                                                                                          |
| 21 |                               |                    |            |          |            |                                                                                                                                                          |
| 22 |                               |                    |            |          |            |                                                                                                                                                          |
| 23 |                               |                    |            |          |            |                                                                                                                                                          |
| 24 |                               |                    |            |          |            |                                                                                                                                                          |
| 25 |                               |                    |            |          |            |                                                                                                                                                          |

+Raw IntakeRouting TrackerAI Decision Log

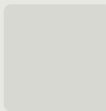
# OPERATIONAL SAFEGUARDS



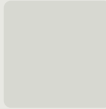
Confidence thresholds prevent incorrect routing



Full audit logging of AI decisions

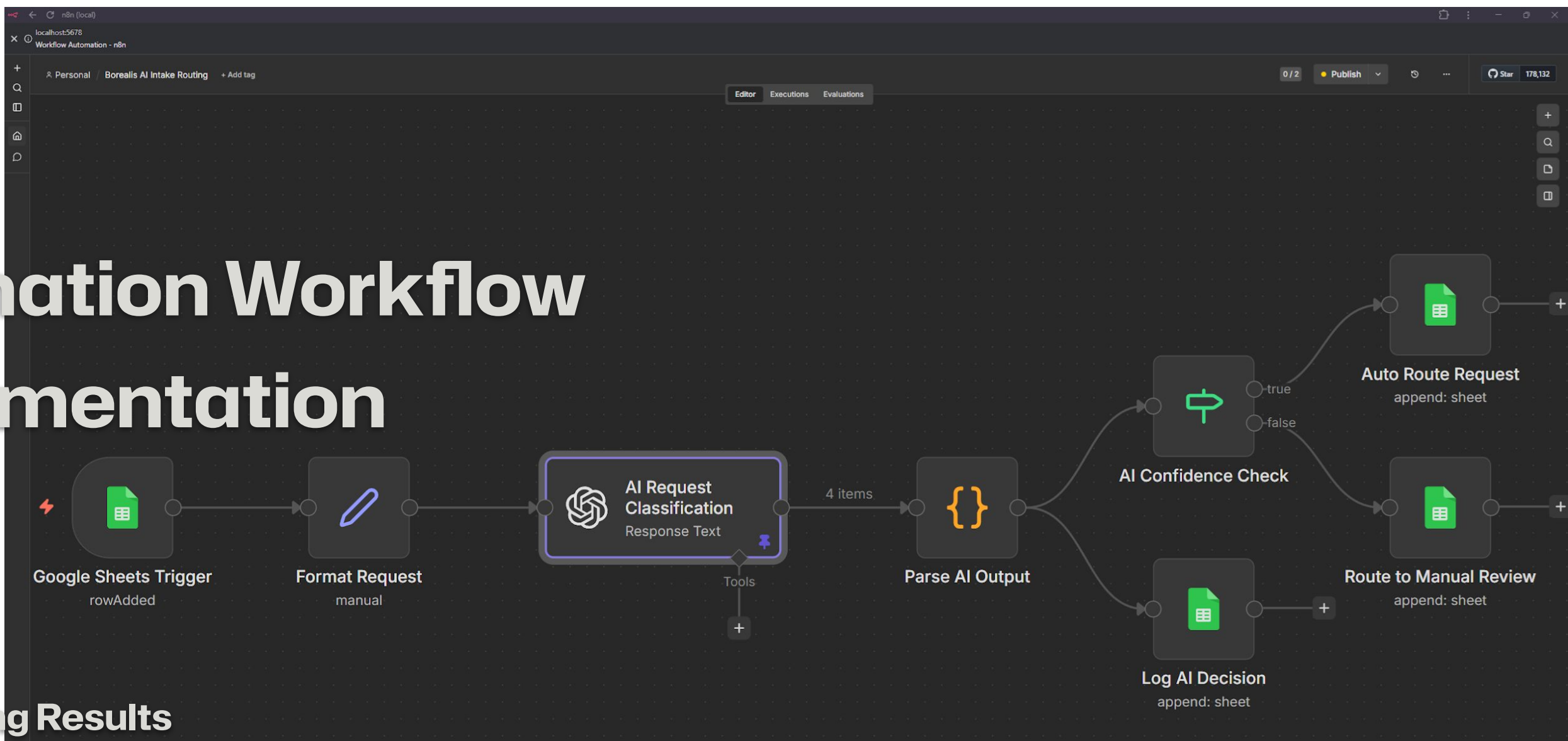


Manual review queue for uncertain classifications



Prompt tuning for continuous improvement

# Automation Workflow Implementation



## Automated Routing Results

| Timestamp                     | Request Type       | Owner      | Priority | Summary                                                            | Recommended Action                                                      | Status          | # | Confidence | Reasoning Note                                                                                      |
|-------------------------------|--------------------|------------|----------|--------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------|---|------------|-----------------------------------------------------------------------------------------------------|
| 2026-03-07T15:43:28.043-05:00 | Marketing Request  | Marketing  | Medium   | Update the spring campaign landing page with ne                    | Execute the update of the landing page as per the                       | Auto Classified |   | 0.95       | The request is clearly related to marketing activities and specifies an update to a landing page, v |
| 2026-03-07T15:43:28.044-05:00 | IT Support         | Operations | High     | Issues connecting to VPN after password reset.                     | Investigate VPN connectivity issues and assist wi                       | Auto Classified |   | 0.95       | The request concerns a technical issue related to VPN access, which falls under IT support, and     |
| 2026-03-07T15:43:28.045-05:00 | HR Request         | HR         | High     | New employee requires access to the payroll syst                   | Grant payroll system access to the new hire imme                        | Auto Classified |   | 0.95       | The request is clear and urgent as it pertains to enabling a new employee's access prior to their   |
| 2026-03-07T15:43:28.046-05:00 | Finance Request    | Finance    | Low      | Inquiry about the approval status of invoice #448:                 | Check the invoice approval system and respond w                         | Auto Classified |   | 0.9        | The request is specifically related to finance operations, focusing on invoice processing and app   |
| 2026-03-07T15:49:04.470-05:00 | Operations Request | Operations | Medium   | Incorrect sales numbers on the marketing dashbc                    | Investigate the reporting issue and data synchron                       | Auto Classified |   | 0.85       | The request involves data accuracy within the dashboard, directly related to operations.            |
| 2026-03-07T15:50:06.170-05:00 | IT Support         | HR         | Medium   | A new hire is facing access issues with the emplo                  | Investigate the new hire's account permissions ar                       | Auto Classified |   | 0.85       | The request relates to technical access problems, which typically fall under IT support, even tho   |
| 2026-03-07T15:50:07.483-05:00 | General / Unclear  | General    | Low      | Request to automate approval process for interna                   | Evaluate existing approval workflows and explore                        | Manual Review   |   | 0.7        | The request is not specific to any department and lacks immediate urgency, making it a general      |
| 2026-03-07T15:51:03.506-05:00 | Operations Request | Operations | Low      | Issue with incorrect numbers in reporting system.                  | Investigate the source of incorrect data.                               | Auto Classified |   | 0.85       | The request involves a reporting issue within the operations department, indicating a need for op   |
| 2026-03-07T15:54:04.200-05:00 | IT Support         | Finance    | Medium   | Access issue with marketing analytics dashboard                    | Investigate access permissions and data accurac                         | Auto Classified |   | 0.85       | The request involves access issues related to IT systems and the potential inaccuracy of data, w    |
| 2026-03-08T16:22:32.350-04:00 | Marketing Request  | Marketing  | High     | Update pricing on spring promotion page to reflect current offers. | Update the pricing table and call-to-action button on the landing page. | Auto Classified |   | 0.95       | The request pertains specifically to marketing activities and is time-sensitive due to an upcomin   |



# EXPECTED IMPACT

## **SIGNIFICANT REDUCTION IN MANUAL TRIAGE**

Less time spent on request classification and routing

## **ACCELERATED REQUEST PROCESSING**

High-priority requests routed immediately upon classification

## **STANDARDIZED CLASSIFICATION**

Uniform request categorization across all departments

By automating the classification and routing process, teams can focus on resolution rather than triage. Metrics would be measured during pilot deployment.

# OPERATIONAL VALUE & ROI POTENTIAL

## TIME SAVINGS

- Average manual request triage time: ~3 minutes per request
- Automated classification time: ~2 seconds per request
- Example request volume: ~500 requests per month
- Estimated time saved: ~25 hours per month (~300 hours annually)

## COST REDUCTION

- Estimated operations analyst cost: ~\$40/hour
- Monthly labor savings potential: ~\$1,000
- Annual labor savings potential: ~\$12,000

## OPERATIONAL BENEFITS

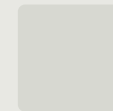
- Faster response times for internal teams
- Reduced routing errors through automated classification
- Consistent request categorization across departments
- Scalable request handling during high-volume periods

*Illustrative estimate based on assumed request volume and average manual triage time.*

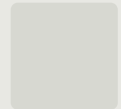
# FUTURE IMPROVEMENTS



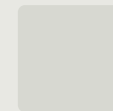
Automated notifications to department owners when requests are routed



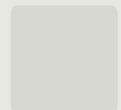
Integration with enterprise ticketing systems such as Jira or ServiceNow



Dashboard analytics to track request trends and department workload



Adaptive confidence thresholds based on historical accuracy



Prompt refinement or model tuning to improve classification accuracy



# IMPLEMENTATION ROADMAP

## PHASE 1 – PILOT DEPLOYMENT *(2-4 weeks)*

- Deploy the automation workflow for an initial department pilot
- Monitor AI classification accuracy and routing outcomes
- Tune prompts and confidence thresholds for reliability

## PHASE 2 – OPERATIONAL EXPANSION *(4-8 weeks)*

- Expand intake to additional departments and request types
- Introduce automated notifications for routed requests
- Implement dashboard analytics to track request volume and routing patterns

## PHASE 3 – ENTERPRISE INTEGRATION *(2-3 months)*

- Integrate with enterprise ticketing systems such as Jira or ServiceNow
- Implement authentication, access controls, and production monitoring
- Scale the automation workflow across organizational operations